

Science of Retrieval - Vector Space Model

Information Retrieval

Sean MacAvaney

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Objectives

- Discuss the science of retrieval
- Is there more to it than simple implementation (or counting!)?
 - Vector space model
 - A historic model based on the **vector-space theory**
 - Still a popular approach for implementation:
Default in Apache Lucene
 - It can incorporate advanced weighting schemes
- Looking at the Library Systems
 - ...for intuition

What is the intuition behind the Vector Space Model?

(Recall) Bag of Words Representation

- Simple strategy for **representing documents**
- **Count** how many times each term occurs
 - Binary mode uses only 0 & 1
- A '**term**' is any lexical item that you chose such as:
 - A word (delimited by 'white space' or punctuation)
 - Some conflated 'root form' of each word (e.g. a stem)
 - An **n-gram** (a sequence of any consecutive n chars)
- Doesn't consider the **ordering of words** in a document
 - **John is quicker than Mary** and **Mary is quicker than John** have the same representation
 - This could be a setback: **positional information** and **neural network architectures** allow us to distinguish these two docs
- For now: Bag of Words Model (BoW)

Vector Space Model

**LET'S LOOK AT THIS PROCESS
DIFFERENTLY**

Document Vectors

One location for each word

	diet	film	fur	galaxy	heat	h'wood	nova	role
A				5	3		10	

“Nova” occurs 10 times in text A

“Galaxy” occurs 5 times in text A

“Heat” occurs 3 times in text A

(Blank means 0 occurrences.)

Document Vectors

One location for each word

diet film fur galaxy heat h'wood nova role

“Hollywood” occurs 7 times in text I
“Film” occurs 5 times in text I
“Diet” occurs 1 time in text I
“Fur” occurs 3 times in text I

I	1	5	3			7		
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Document Vectors

One vector for each document

Document ids

↓	diet	film	fur	galaxy	heat	h'wood	nova	role
A				5	3		10	
B	5	10						
C				10	8	7		
D				9	10	5		
E							10	10
F							9	10
G	5	7			9			
H		6	10	2	8			
I	1	5	3			7		

Vector Space Model

- Documents are also treated as a “bag” of words or terms
- Each document is represented as a vector in a *t-dimensional* vector space (t is the number of index terms)
- Each term weight is computed based on some variations of **TF** or **TF-IDF** scheme

TF-IDF Vectors

Document ids

↓	diet	film	fur	galaxy	heat	h'wood	nova	role
A				8	.5		.6	
B	4	1						
C				3	4	2.8		
D				2.7	5	2		
E							9	1.5
F							8.1	1.5
G	4	2.8			3.6			
H		.6	4		1			
I	2.1	2.5	.9				.45	

Sparse Matrix

More Formally

- Documents and queries are represented by **vectors** of term weights
- A collection is represented by a matrix of term weights

$$D_i = (d_{i1}, d_{i2}, \dots, d_{it}) \quad Q = (q_1, q_2, \dots, q_t)$$

	<i>Term</i> ₁	<i>Term</i> ₂	...	<i>Term</i> _{<i>t</i>}
<i>Doc</i> ₁	<i>d</i> ₁₁	<i>d</i> ₁₂	...	<i>d</i> _{1<i>t</i>}
<i>Doc</i> ₂	<i>d</i> ₂₁	<i>d</i> ₂₂	...	<i>d</i> _{2<i>t</i>}
⋮	⋮			
<i>Doc</i> _{<i>n</i>}	<i>d</i> _{<i>n</i>1}	<i>d</i> _{<i>n</i>2}	...	<i>d</i> _{<i>n</i><i>t</i>}

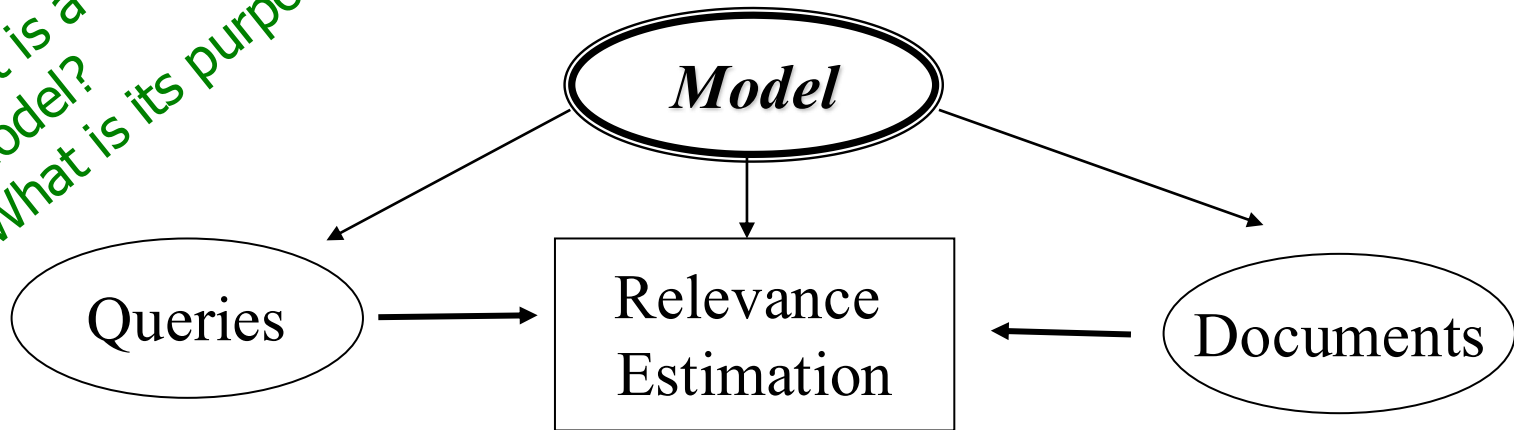
So, we have docs represented as vectors

How can we use this in retrieval?

t is the number of index terms (words, stems, etc)

Retrieval Model

What is a retrieval model?
What is its purpose?



An **IR Model** defines

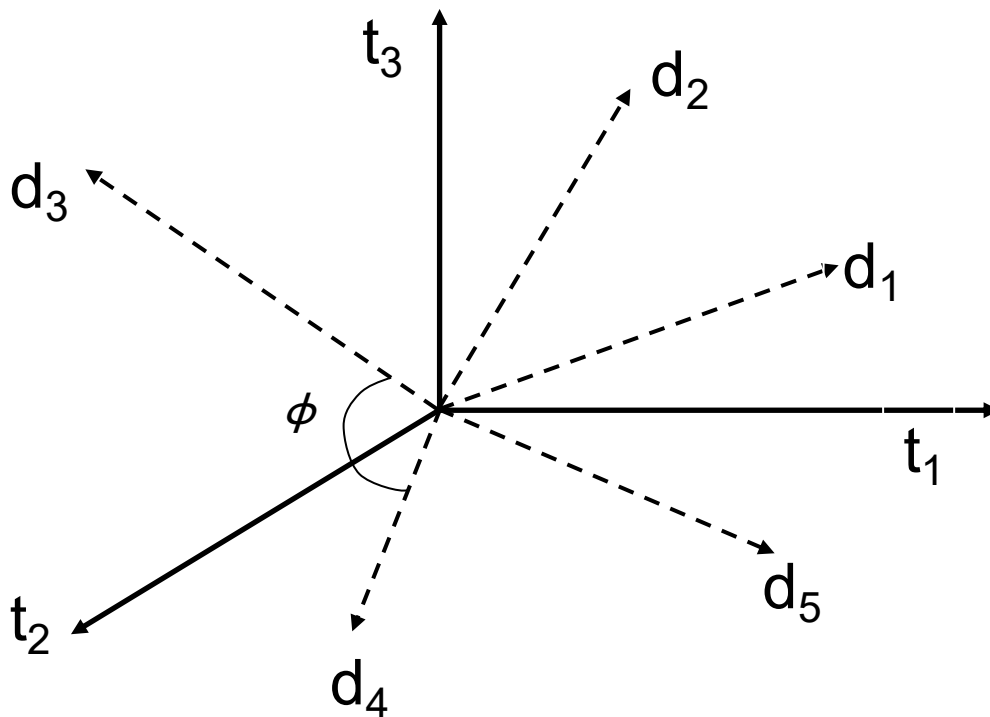
- a model for document representation
- a model for query representation
- a mechanism for estimating the relevance of a document for a given query

Progress in retrieval models has corresponded with improvements in effectiveness

Retrieval in Vector Space Model

- Vector space model represents both query and documents using term sets (**term vectors**)
- Documents and queries are represented in a high dimensional space (Bag of Words)
 - Each dimension of the space corresponds to a term in the document collection (*t-dimensional vector space*)
- **Relevance Estimation** is performed by identifying documents **similar** to the query
 - **Relevance of $\mathbf{d}_i$ to $\mathbf{q}$** : Compare the **similarity** of query $\mathbf{q}$ and document $\mathbf{d}_i$

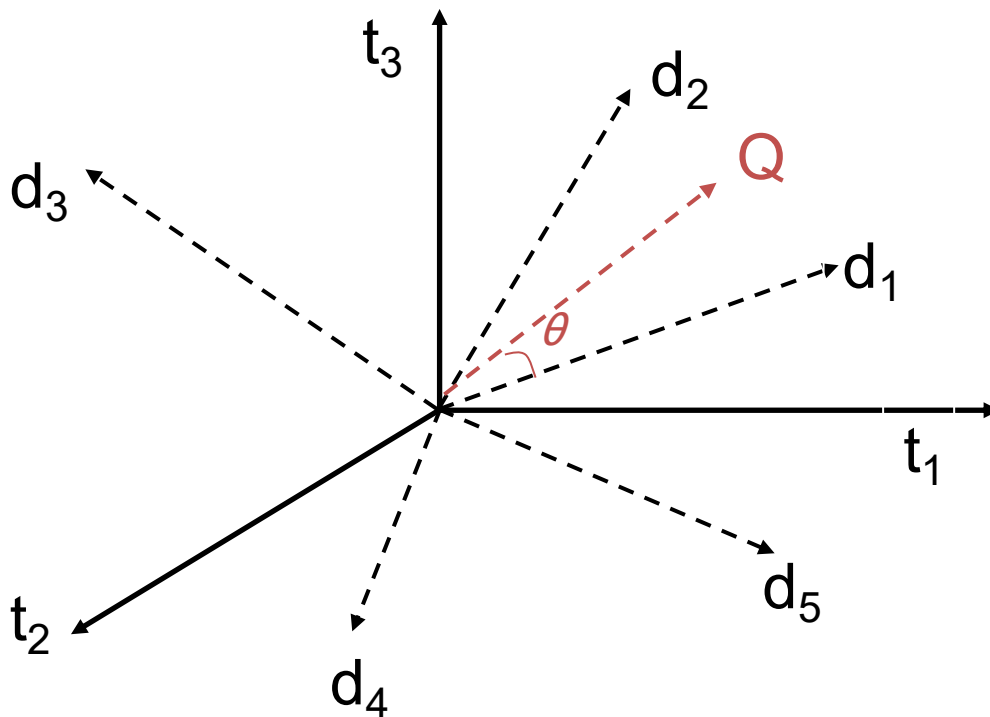
Geometrically: Vector Space Model



Assumption: Documents that are “close together” in vector space “talk about” the same things

NB: 3D diagrams useful, but can be misleading for high-dimensional space

Geometrically: Vector Space Model



Assumption: Documents that are “close together” in vector space “talk about” the same things

Therefore, retrieve documents based on **how close the document is to the query** (i.e., similarity $\sim$ “closeness”)

Vector Space

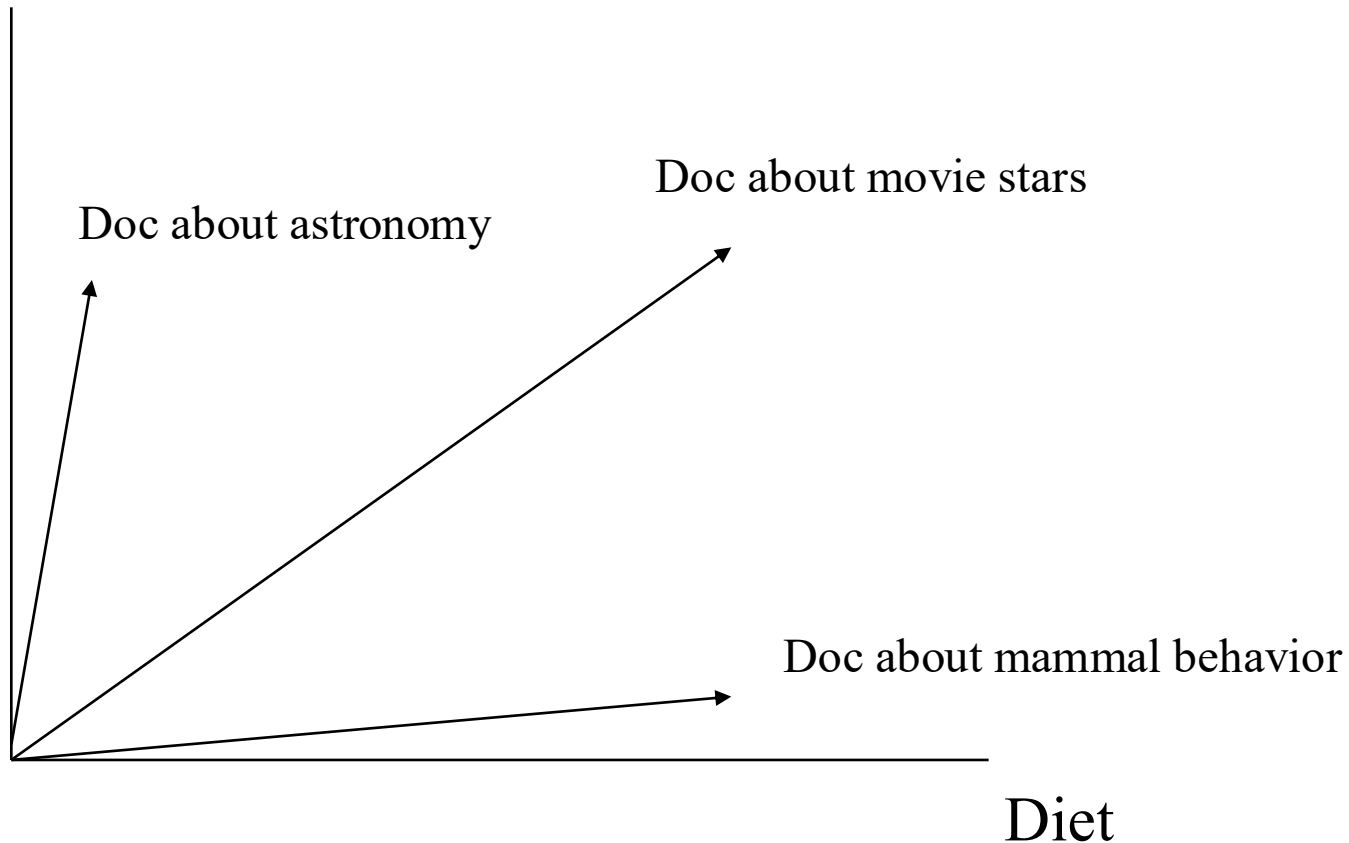
- $X = (t_1, t_2, \dots, t_t)$
 - The number t_i is called the **i**-th component of the vector
 - **Magnitude:** is defined by the square root of the sum of the squares of the components
 - that is, $\sqrt{\sum t_i^2}$
 - If $\|X\| = 1$ then X is a **unit vector**
 - *Concept of length normalization*

Summary: Document Vectors

- Documents are represented as “bags of words”
- Represented as vectors when used computationally
 - A vector is like an array of floating point
 - Has direction and magnitude
 - Each vector holds a (unique) place for **every term** in the collection
 - Therefore, most vectors are **sparse**

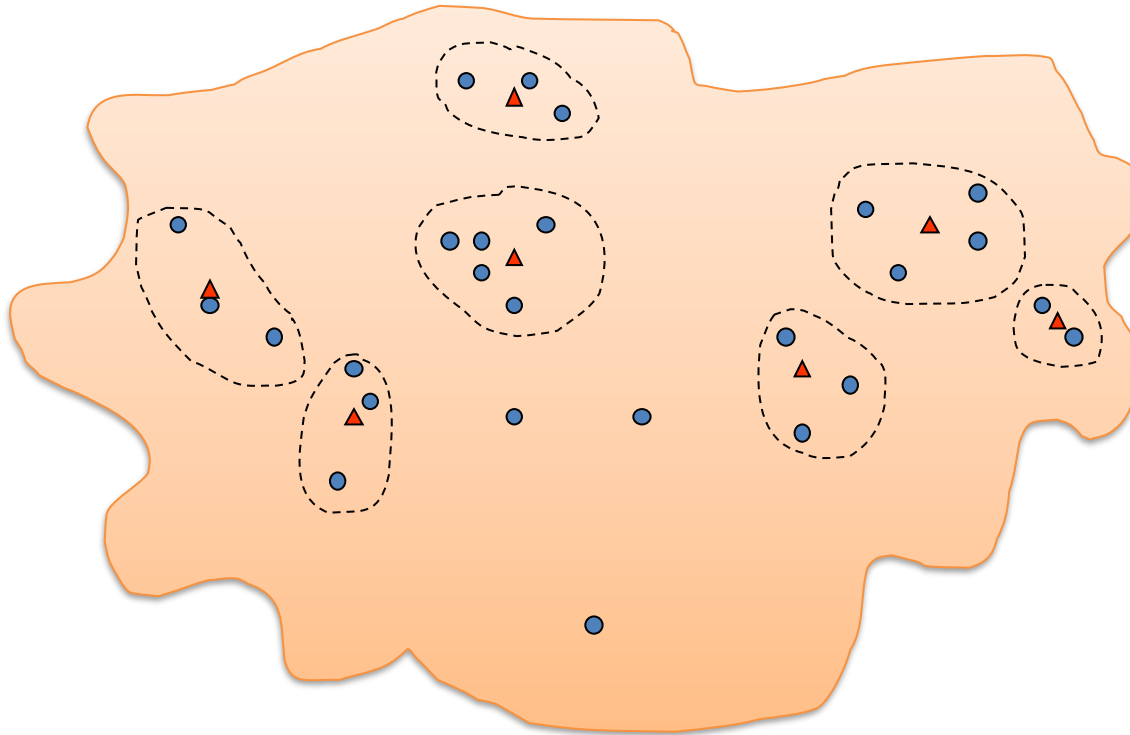
Plotting the Vectors ... & Intuition

Star



Vector Space Intuition

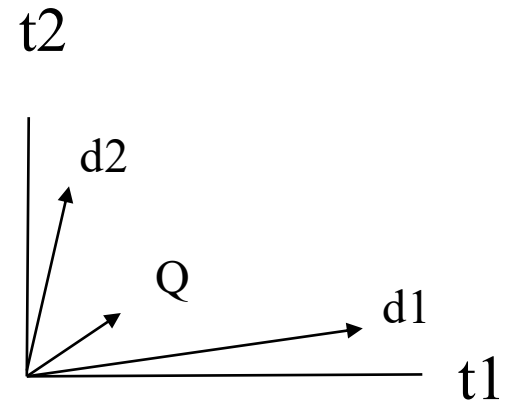
- Library
 - Books from a domain are organised at the same place/ shelf/ nearby shelves
 - Human organisation - librarian



What is the intuition behind
The vector-space model?

Vector Space Model

- The relevant documents for a query are expected to be those represented by the vectors closest to the query



- Documents ranked by distance between points representing query and documents
 - **Similarity** measure more common than a distance or *dissimilarity* measure

– e.g.

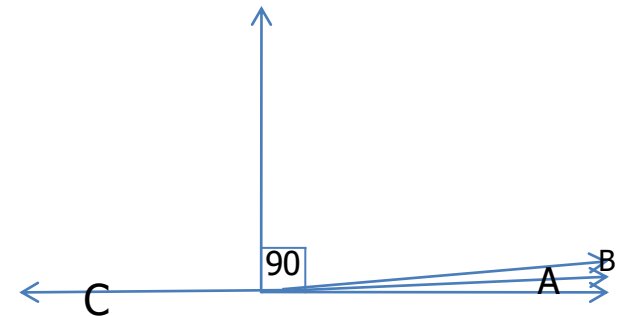
$$\text{Cosine}(D_i, Q) = \frac{\sum_{j=1}^t d_{ij} \cdot q_j}{\sqrt{\sum_{j=1}^t d_{ij}^2 \cdot \sum_{j=1}^t q_j^2}}$$

Cosine Measure

In IR we consider only the similarity range from 0 to 1
Why? Why not -1 to 1?

- It measures cosine of the angle between the vectors
- Cosine ranges from 1 for vectors pointing in the same direction over zero for orthogonal vectors and -1 for vectors pointing in opposite directions
- If Cosine is applied to normalised (unit) vectors it gives the same ranking as **Euclidean distance** does

$$\begin{aligned}\cos 0^\circ &= 1 & \cos 90^\circ &= 0 \\ \cos 180^\circ &= -1\end{aligned}$$



Similarity Calculation

- Consider two documents D_1, D_2 and a query Q
 - $D_1 = (0.5, 0.8, 0.3), D_2 = (0.9, 0.4, 0.2), Q = (1.5, 1.0, 0)$



WITHELD

How could we implement a cosine similarity-based measure using inverted index?

What role the denominator plays?

Algorithm (Reminder)

For each document I, $\text{Score}(I) = 0$; $I = 1$ to N

For each query term t_k

- Search the vocabulary list
- Pull out the postings list
- For each document J in the list,
 - $\text{Score}(J) = \text{Score}(J) + w_{kj}$

Example

- $D1 = (T1 \Rightarrow 12, T2 \Rightarrow 23, T3 \Rightarrow 3)$
- $D2 = (T1 \Rightarrow 3, T2 \Rightarrow 2, T3 \Rightarrow 1)$
- $Q = (T1 \Rightarrow 0, T2 \Rightarrow 0, T3 \Rightarrow 2)$

- $\text{Sim}(D1, Q) = 12 * 0 + 23 * 0 + 3 * 2 = 6$

- $\text{Sim}(D2, Q) = 3 * 0 + 2 * 0 + 1 * 2 = 2$

If you are using an inverted index?
 $3 * 2 = 6$
 $1 * 2 = 2$
Role of Document length?

Matching Coefficient (Coordination Level)

- Simply counts the number of dimensions on which both vectors are non-zero
- $|X \cap Y| \equiv \sum x_i * y_i$
- Number of shared index terms (binary vectors)
- Does not take into account the sizes of the vectors

Is this familiar?

Some Problems ...

- Normalisation ...
- Consider a single word query and a single word document (**In Binary mode...**)
 - If that matches
 - Coefficient is 1
- Same query against a thousand word document
 - If that matches
 - Coefficient is 1

Justify the need for vector
length normalization

Dice Coefficient

- $2 |X \cap Y| / (|X| + |Y|)$
- Normalises for length by dividing by the total number of non-zero entries.
- We multiply by 2 so that we get a measure that ranges from 0 to 1.0

What is a dice coefficient?

Query Term Weighting

- Boolean representation
 - Just have a weight of zero or 1
- Short queries
 - Typical of web searches
 - Multiple keyword occurrences are rare
 - $W_{kq} = \text{idf}_k$
- Long queries
 - Result of **relevance feedback** (will talk about it later)
 - $W_{kq} = f_{kq} \cdot \text{idf}_k$

Discuss three query term weighting strategies!

Advantages and Disadvantages of a Vector-space Model

Advantages

- Simple **geometric interpretation** of retrieval readily comprehensible to non-specialist and a uniform basis for wide range of operations
- Easy to compute measure (any similarity measure can be used)
- Easy to adapt to various weighting schemes
- Provision for *ranked output*

Disadvantages

- High dimensionality
- Term independence assumption
- Adhoc similarity metric: Cosine, Dice, etc. (which one to use?)
- Adhoc term weighting (not theoretically founded)
- No guidance on when to stop ranking